LAB 02 / NETWORK ROUTING

Multi-Area OSPF Routing

Cisco router configuration and adjacency troubleshooting

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Executive Summary

Documented a multi-area OSPF exercise using Cisco routers and switches. The work covers interface addressing, router IDs, area assignments, and troubleshooting missing neighbor relationships caused by mistyped addresses.

SKILLS DEMONSTRATED

OSPF · IPv4 and IPv6 · Cisco IOS · PuTTY · Packet Tracer

- Organized a topology around backbone and non-backbone OSPF areas.
- Recorded IPv4 and IPv6 interface configuration and router IDs.
- Identified incorrect interface addresses while troubleshooting missing neighbors.

Evidence Included

Annotated topology, per-device configuration screenshots, route information, and OSPF process output.

The reference configurations are reconciled with the original IPv4 and IPv6 topology diagrams. Original console screenshots remain labeled as historical evidence. Corrected examples were checked for addressing and area consistency, without a new device run.

Objective

The objective was to configure and troubleshoot a multi-area OSPF network using Cisco routers and switches. PuTTY provided command-line access, while Packet Tracer supported topology planning and visualization.

Technical Background

Open Shortest Path First (OSPF) is a link-state interior gateway protocol. Routers exchange link-state information and calculate routes using a shortest-path algorithm. Neighbor formation, consistent interface addressing, and compatible area settings are essential to operation.

Multi-area OSPF divides routing information into areas connected through the backbone, Area 0. An area border router (ABR) connects multiple areas. This structure can reduce the amount of topology information processed within each area and support larger networks.

PuTTY is a terminal client used to access device command-line interfaces. In this exercise, it was used to configure interfaces, routing processes, router IDs, and verification commands.

Cisco routers and switches form the lab topology. Routed interfaces connect network segments, and OSPF exchanges route information between participating devices. Both IPv4 and IPv6 addressing appear in the configuration record.

Procedure and Evidence

The lab covers interface addressing, OSPF areas, and router IDs. The corrected reference configurations below follow the original topology diagrams. Original device screenshots are retained separately and may reflect earlier configuration states.

Configuration Record

Network Topology

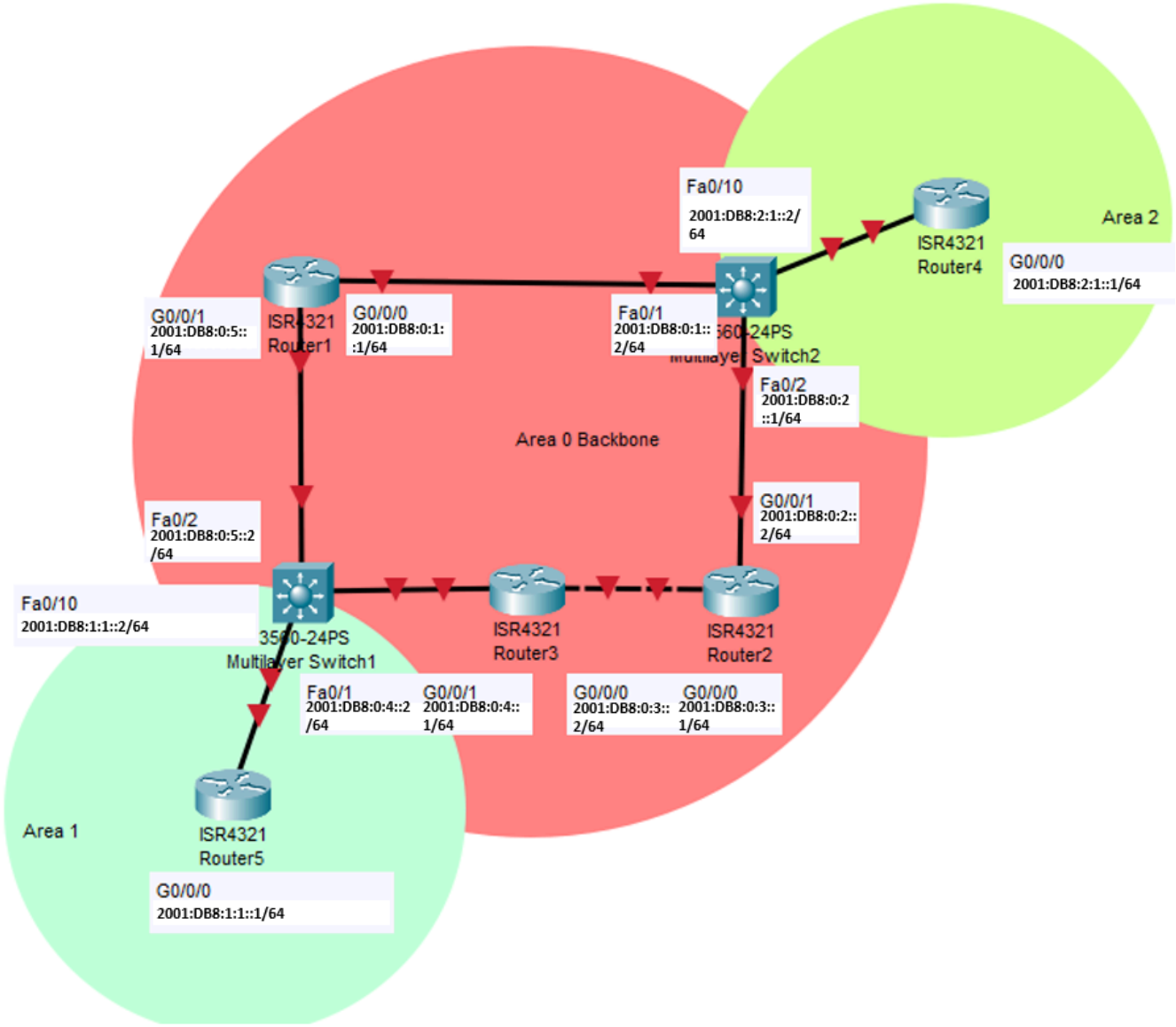


Figure 01. Network Topology

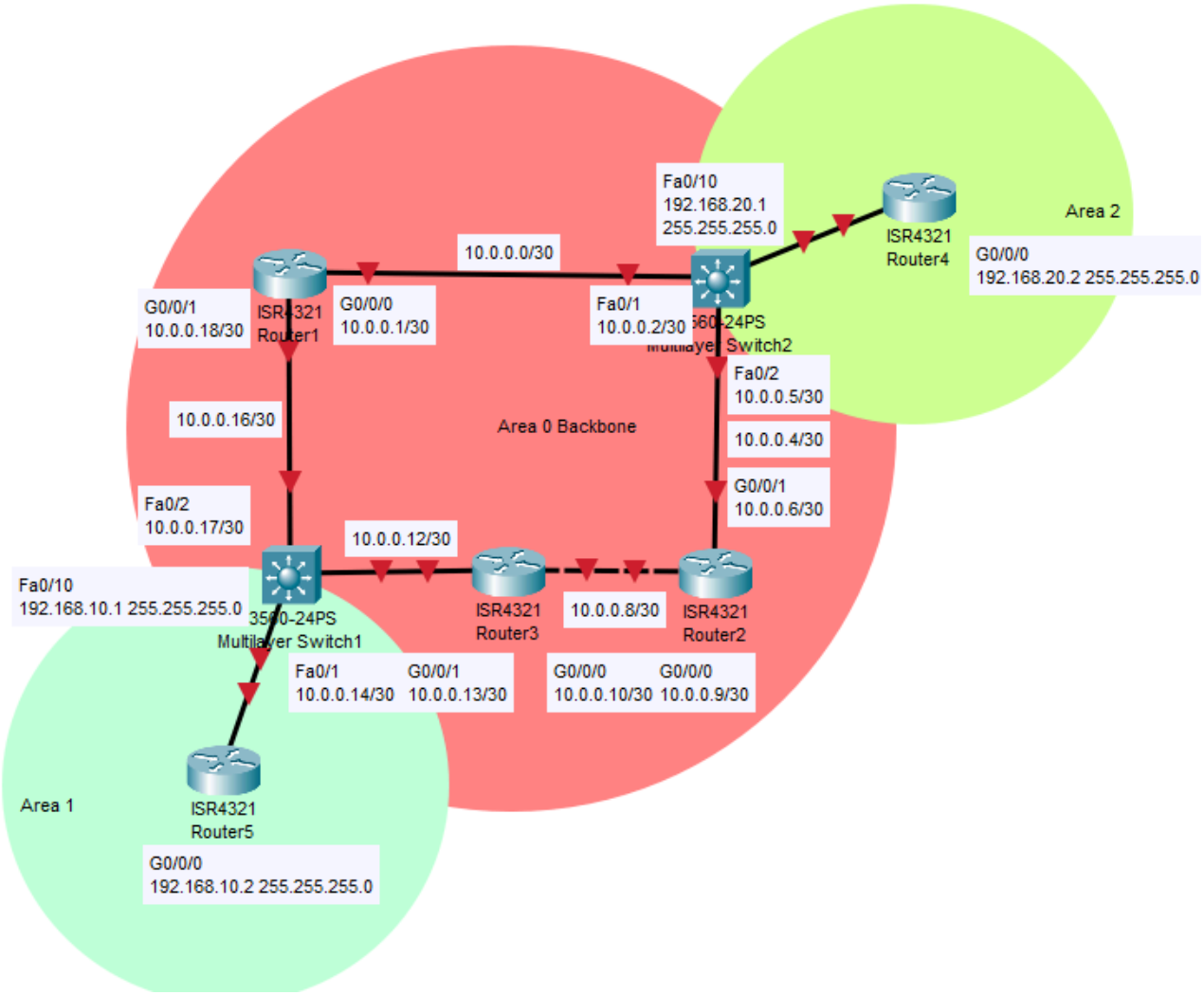


Figure 02. Network Topology

Corrected Addressing Reference

The table below follows the original topology diagrams: five backbone links in Area 0, S1–R5 in Area 1, and S2–R4 in Area 2. Router IDs are unique. Use a clean lab configuration and IOS images supporting routed switch ports and classic OSPFv3 commands; interface names and IPv6 routing support depend on the platform.

Device / Interface	IPv4 / Prefix	IPv6 / Prefix	Area
R1 g0/0/0	10.0.0.1/30	2001:db8:0:1::1/64	0
S2 fa0/1	10.0.0.2/30	2001:db8:0:1::2/64	0
S2 fa0/2	10.0.0.5/30	2001:db8:0:2::1/64	0
R2 g0/0/1	10.0.0.6/30	2001:db8:0:2::2/64	0
R2 g0/0/0	10.0.0.9/30	2001:db8:0:3::1/64	0

Device / Interface	IPv4 / Prefix	IPv6 / Prefix	Area
R3 g0/0/0	10.0.0.10/30	2001:db8:0:3::2/64	0
R3 g0/0/1	10.0.0.13/30	2001:db8:0:4::1/64	0
S1 fa0/1	10.0.0.14/30	2001:db8:0:4::2/64	0
S1 fa0/2	10.0.0.17/30	2001:db8:0:5::2/64	0
R1 g0/0/1	10.0.0.18/30	2001:db8:0:5::1/64	0
S1 fa0/10	192.168.10.1/24	2001:db8:1:1::2/64	1
R5 g0/0/0	192.168.10.2/24	2001:db8:1:1::1/64	1
S2 fa0/10	192.168.20.1/24	2001:db8:2:1::2/64	2
R4 g0/0/0	192.168.20.2/24	2001:db8:2:1::1/64	2

Per-interface OSPF activation replaces the mismatched network statements. Both OSPFv2 and OSPFv3 have explicit router IDs. These are corrected reference examples, not a transcript of the retained screenshots.

R1: Corrected Reference Configuration

```
enable
configure terminal
hostname R1
ipv6 unicast-routing
router ospf 1
 router-id 1.1.1.1
 exit
ipv6 router ospf 1
 router-id 1.1.1.1
 exit
interface g0/0/0
 description Link to S2 fa0/1
 ip address 10.0.0.1 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:1::1/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface g0/0/1
 description Link to S1 fa0/2
 ip address 10.0.0.18 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:5::1/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
end
show running-config
```

```
show ip ospf neighbor
show ipv6 ospf neighbor
```

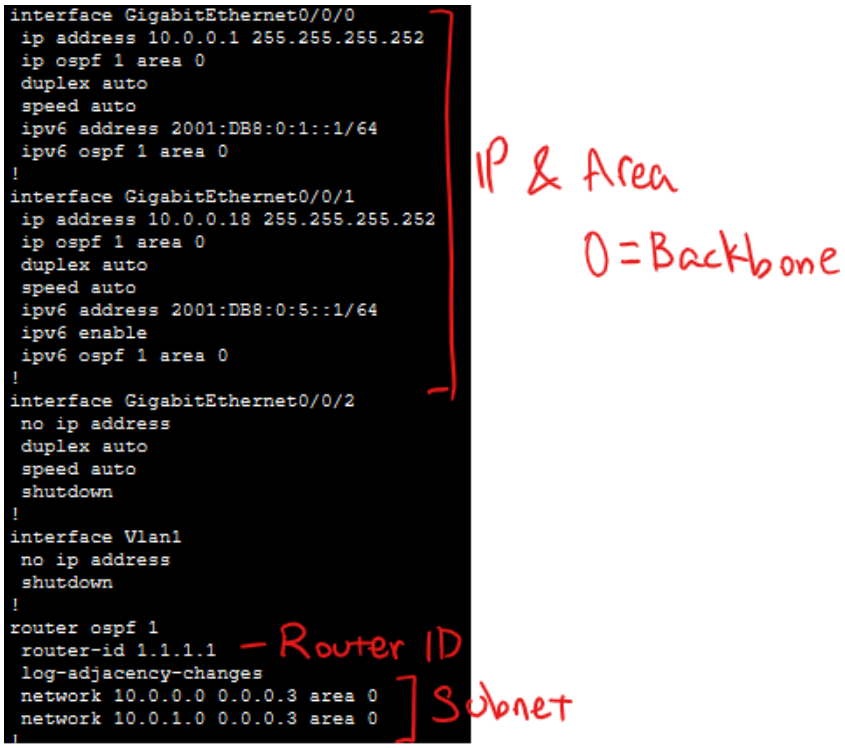


Figure 03. Original lab capture for R1; separate from the corrected reference configuration.

R2: Corrected Reference Configuration

```
enable
configure terminal
hostname R2
ipv6 unicast-routing
router ospf 1
 router-id 2.2.2.2
 exit
ipv6 router ospf 1
 router-id 2.2.2.2
 exit
interface g0/0/1
 description Link to S2 fa0/2
 ip address 10.0.0.6 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:2::2/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface g0/0/0
 description Link to R3 g0/0/0
 ip address 10.0.0.9 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:3::1/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
```

```
end
show running-config
show ip ospf neighbor
show ipv6 ospf neighbor
```

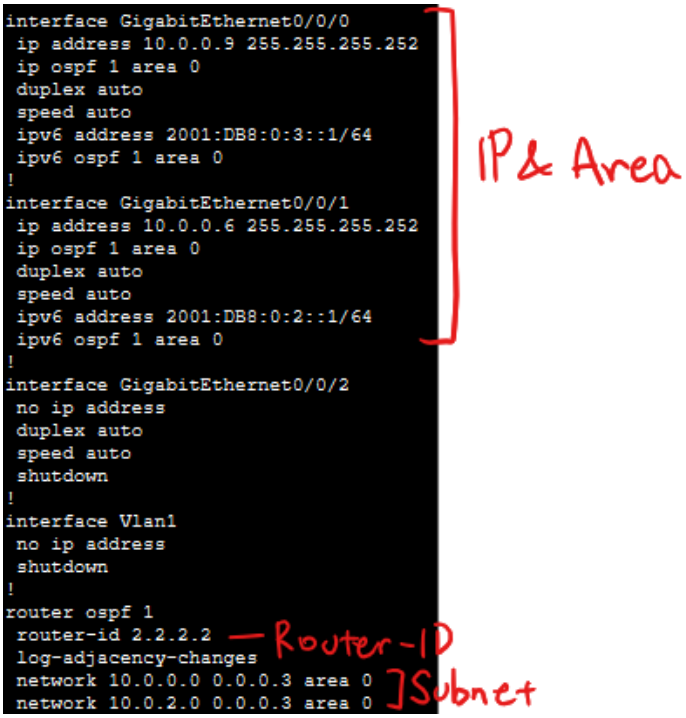


Figure 04. Original lab capture for R2; separate from the corrected reference configuration.

R3: Corrected Reference Configuration

```
enable
configure terminal
hostname R3
ipv6 unicast-routing
router ospf 1
 router-id 3.3.3.3
 exit
ipv6 router ospf 1
 router-id 3.3.3.3
 exit
interface g0/0/0
 description Link to R2 g0/0/0
 ip address 10.0.0.10 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:3::2/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface g0/0/1
 description Link to S1 fa0/1
 ip address 10.0.0.13 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:4::1/64
 ipv6 ospf 1 area 0
 no shutdown
```

```
 exit
end
show running-config
show ip ospf neighbor
show ipv6 ospf neighbor
```

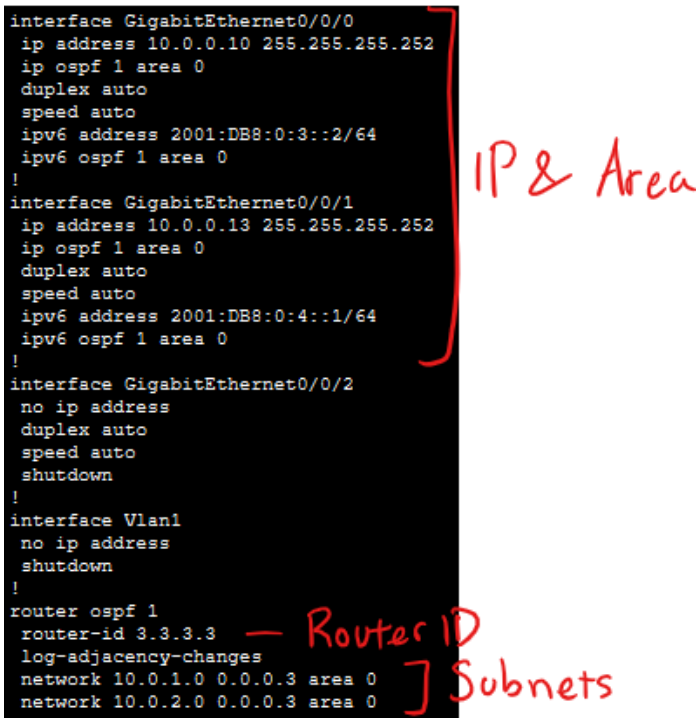


Figure 05. Original lab capture for R3; separate from the corrected reference configuration.

S1: Corrected Reference Configuration

```
enable
configure terminal
hostname S1
ip routing
ipv6 unicast-routing
router ospf 1
 router-id 10.10.10.10
 exit
ipv6 router ospf 1
 router-id 10.10.10.10
 exit
interface fa0/1
 description Link to R3 g0/0/1
 no switchport
 ip address 10.0.0.14 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:4::2/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface fa0/2
 description Link to R1 g0/0/1
 no switchport
 ip address 10.0.0.17 255.255.255.252
```

```
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:5::2/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface fa0/10
 description Link to R5 g0/0/0
 no switchport
 ip address 192.168.10.1 255.255.255.0
 ip ospf 1 area 1
 ipv6 address 2001:db8:1:1::2/64
 ipv6 ospf 1 area 1
 no shutdown
 exit
end
show running-config
show ip ospf neighbor
show ipv6 ospf neighbor
```

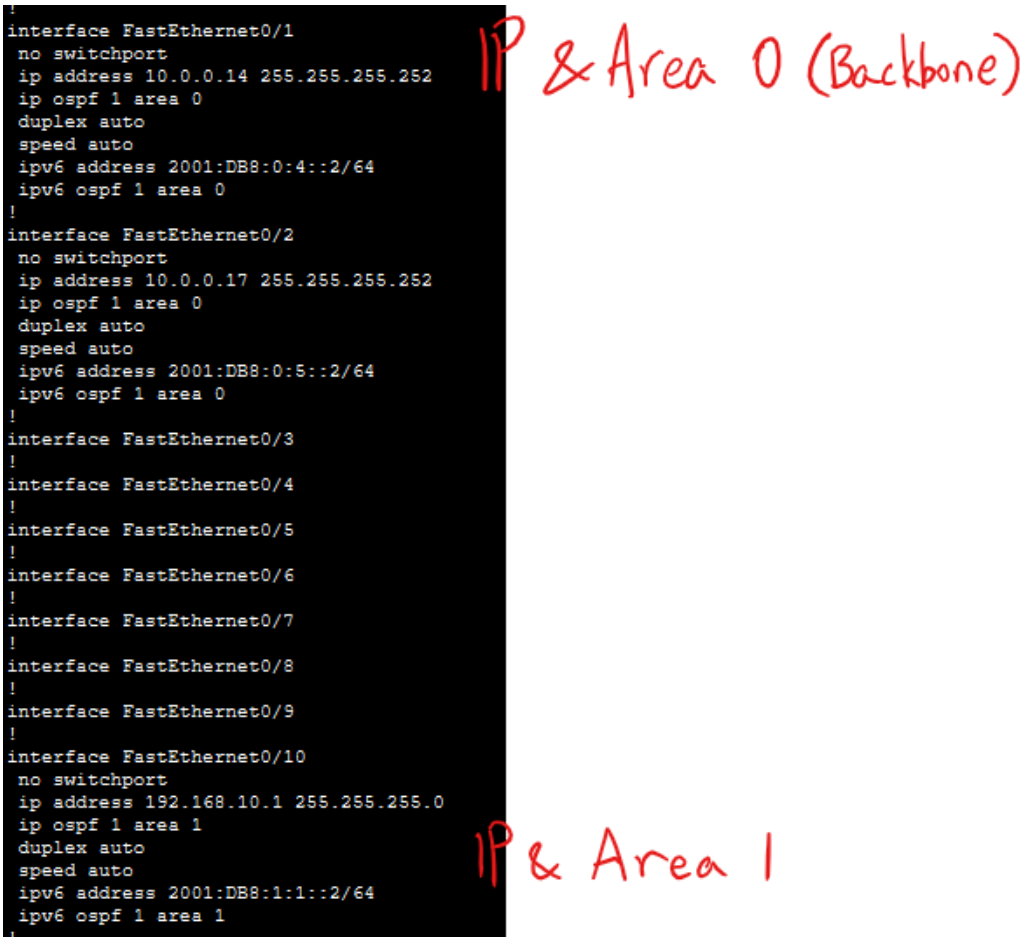


Figure 06. Original lab capture for S1; separate from the corrected reference configuration.

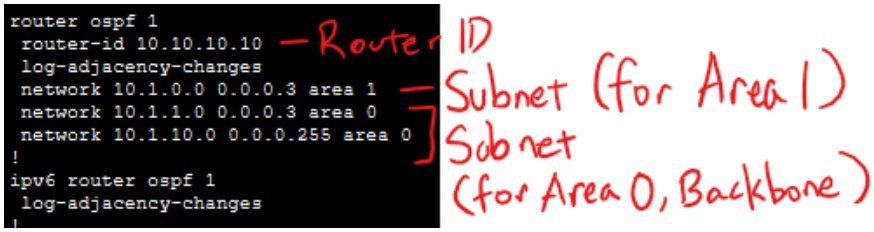


Figure 07. Original lab capture for S1; separate from the corrected reference configuration.

S2: Corrected Reference Configuration

```
enable
configure terminal
hostname S2
ip routing
ipv6 unicast-routing
router ospf 1
 router-id 11.11.11.11
 exit
ipv6 router ospf 1
 router-id 11.11.11.11
 exit
interface fa0/1
 description Link to R1 g0/0/0
 no switchport
 ip address 10.0.0.2 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:1::2/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface fa0/2
 description Link to R2 g0/0/1
 no switchport
 ip address 10.0.0.5 255.255.255.252
 ip ospf 1 area 0
 ipv6 address 2001:db8:0:2::1/64
 ipv6 ospf 1 area 0
 no shutdown
 exit
interface fa0/10
 description Link to R4 g0/0/0
 no switchport
 ip address 192.168.20.1 255.255.255.0
 ip ospf 1 area 2
 ipv6 address 2001:db8:2:1::2/64
 ipv6 ospf 1 area 2
 no shutdown
 exit
end
show running-config
show ip ospf neighbor
show ipv6 ospf neighbor
```

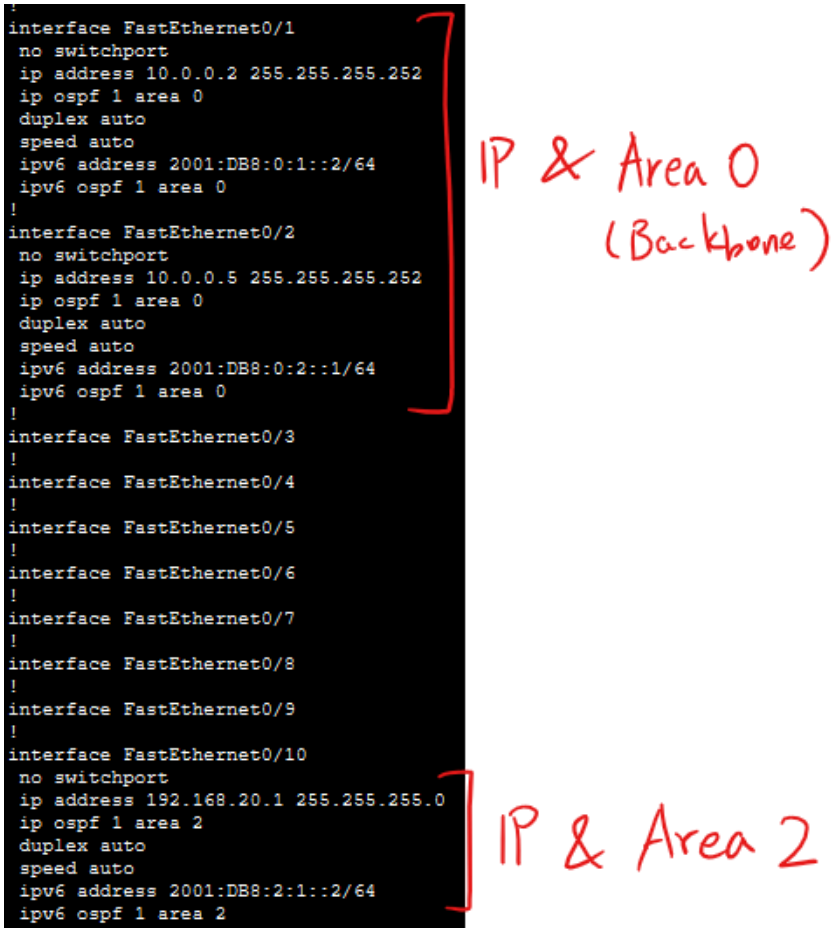


Figure 08. Original lab capture for S2; separate from the corrected reference configuration.

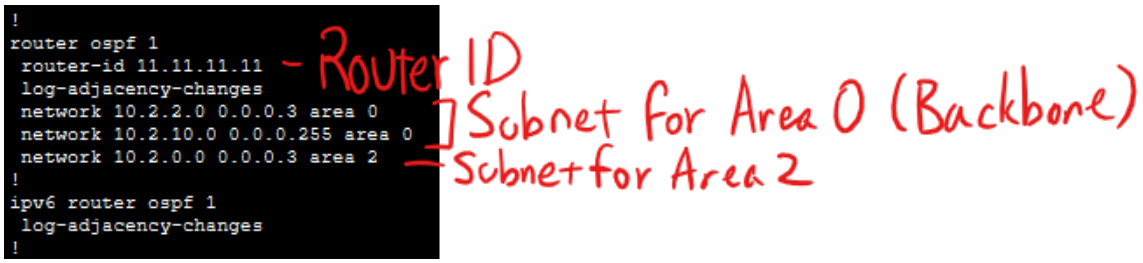


Figure 09. Original lab capture for S2; separate from the corrected reference configuration.

R4: Corrected Reference Configuration

```
enable
configure terminal
hostname R4
ipv6 unicast-routing
router ospf 1
 router-id 4.4.4.4
 exit
ipv6 router ospf 1
 router-id 4.4.4.4
 exit
interface g0/0/0
 description Link to S2 fa0/10
```

```
 ip address 192.168.20.2 255.255.255.0
 ip ospf 1 area 2
 ipv6 address 2001:db8:2:1::1/64
 ipv6 ospf 1 area 2
 no shutdown
 exit
end
show running-config
show ip ospf neighbor
show ipv6 ospf neighbor
```



Figure 10. Original lab capture for R4; separate from the corrected reference configuration.

R5: Corrected Reference Configuration

```
enable
configure terminal
hostname R5
ipv6 unicast-routing
router ospf 1
 router-id 5.5.5.5
 exit
ipv6 router ospf 1
 router-id 5.5.5.5
 exit
interface g0/0/0
 description Link to S1 fa0/10
 ip address 192.168.10.2 255.255.255.0
 ip ospf 1 area 1
 ipv6 address 2001:db8:1:1::1/64
 ipv6 ospf 1 area 1
 no shutdown
 exit
```

```
end
show running-config
show ip ospf neighbor
show ipv6 ospf neighbor
```



Figure 11. Original lab capture for R5; separate from the corrected reference configuration.

Verification Commands and Original Routing Output

Use these IOS verification commands after applying the reference configuration. When the topology converges, the diagram predicts two neighbors each for R1, R2, and R3, three each for S1 and S2, and one each for R4 and R5. These are expected counts, not newly measured results.

```
show ip interface brief
show ipv6 interface brief
show ip ospf neighbor
show ip route ospf
show ip ospf
show ipv6 ospf neighbor
show ipv6 route ospf
show ipv6 ospf
```

From R1, test IPv4 reachability to 192.168.10.2 and 192.168.20.2, and IPv6 reachability to 2001:db8:1:1::1 and 2001:db8:2:1::1. Inspect learned inter-area routes and resolve missing neighbors before claiming convergence. The following images are original lab output.

Router 1

```
router1#show ip ospf route

            OSPF Router with ID (1.1.1.1) (Process ID 1)


                Base Topology (MTID 0)


    Area BACKBONE(0)

    Intra-area Route List
*   10.0.0.0/30, Intra, cost 1, area 0, Connected
      via 10.0.0.1, GigabitEthernet0/0

    Intra-area Router Path List
i 2.2.2.2 [1] via 10.0.0.2, GigabitEthernet0/0, ABR, Area 0, SPF 2

    Inter-area Route List
*>  10.0.0.12/30, Inter, cost 2, area 0
      via 10.0.0.2, GigabitEthernet0/0
*>  10.0.0.16/30, Inter, cost 3, area 0
      via 10.0.0.2, GigabitEthernet0/0

    Area 1

    Intra-area Route List
*   10.0.0.4/30, Intra, cost 1, area 1, Connected
      via 10.0.0.5, GigabitEthernet0/1
*>  10.0.0.8/30, Intra, cost 2, area 1
      via 10.0.0.6, GigabitEthernet0/1
```

Figure 12. Router 1

Router 2

```
r2#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 2.2.2.2
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 It is an area border router
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Retransmission limit dc 24 non-dc 24
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        SPF algorithm executed 2 times
        Number of LSA 10. Checksum Sum 0x064A09
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 2
        Number of interfaces in this area is 1
        SPF algorithm executed 5 times
        Number of LSA 12. Checksum Sum 0x06D4E1
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 13. Router 2

Router 3

```
r3#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 3.3.3.3
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Retransmission limit dc 24 non-dc 24
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area 1
        Number of interfaces in this area is 2
        SPF algorithm executed 5 times
        Number of LSA 14. Checksum Sum 0x0768CF
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 14. Router 3

Router 4

```
r4#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 4.4.4.4
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area 1
        Number of interfaces in this area is 1
        SPF algorithm executed 4 times
        Number of LSA 12. Checksum Sum 0x060651
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 15. Router 4

Router 5

```
r5#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 5.5.5.5
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Retransmission limit dc 24 non-dc 24
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area 2
        Number of interfaces in this area is 2
        SPF algorithm executed 5 times
        Number of LSA 14. Checksum Sum 0x07D4BB
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 16. Router 5

Original OSPF Process Output

Router 1

```
router1#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 1.1.1.1
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 It is an area border router
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        SPF algorithm executed 12 times
        Number of LSA 10. Checksum Sum 0x060B3A
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 1
        Number of interfaces in this area is 1
        SPF algorithm executed 2 times
        Number of LSA 12. Checksum Sum 0x063DDC
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 17. Router 1

Router 2

```
r2#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 2.2.2.2
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 It is an area border router
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Retransmission limit dc 24 non-dc 24
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        SPF algorithm executed 2 times
        Number of LSA 10. Checksum Sum 0x064A09
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 2
        Number of interfaces in this area is 1
        SPF algorithm executed 5 times
        Number of LSA 12. Checksum Sum 0x06D4E1
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 18. Router 2

Router 3

```
r3#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 3.3.3.3
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Retransmission limit dc 24 non-dc 24
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area 1
        Number of interfaces in this area is 2
        SPF algorithm executed 5 times
        Number of LSA 14. Checksum Sum 0x0768CF
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 19. Router 3

Router 4

```
r4#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 4.4.4.4
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area 1
        Number of interfaces in this area is 1
        SPF algorithm executed 4 times
        Number of LSA 12. Checksum Sum 0x060651
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 20. Router 4

Router 5

```
r5#show ospf
 OSPFv3 1 address-family ipv6
 Router ID 5.5.5.5
 Supports NSSA (compatible with RFC 3101)
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Retransmission limit dc 24 non-dc 24
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Graceful restart helper support enabled
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area 2
        Number of interfaces in this area is 2
        SPF algorithm executed 5 times
        Number of LSA 14. Checksum Sum 0x07D4BB
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Figure 21. Router 5

Troubleshooting and Lessons Learned

OSPF neighbors did not initially appear. After reviewing the configuration and discussing the issue with peers, the lab team identified mistyped IP addresses on two routers. Correcting those addresses resolved the reported neighbor issue.

Conclusion

The exercise developed experience with multi-area OSPF design, router IDs, interface addressing, and command-line troubleshooting. It also reinforced the importance of checking IPv4 and IPv6 addresses and validating area membership when adjacencies fail.